🌐 Network Redundancy & Aggregation

This overview merges a Spanning Tree Protocol tutorial that explains loop-prevention in redundant Ethernet designs with an EtherChannel module that details how to build, verify, and troubleshoot link aggregation.

📚 Content Sources

- Section 1: Explains **Spanning Tree** fundamentals for loop prevention in redundant Ethernet networks from a 23-page PDF document. (source)
- Section 2: Describes **EtherChannel** operation, configuration, and troubleshooting of aggregated links from a 35-page PDF document. (source)

Section 1: Spanning Tree Protocol (STP) Fundamentals

This section covers the Spanning Tree Protocol from a 23-page PDF document (source)

Introduction to Redundant Networks and Loops

Modern Ethernet networks implement **redundant paths** using multiple switches and links to ensure fault-tolerant connectivity. While this redundancy protects against single-link failures, it creates the risk of **bridging loops** - situations where frames circulate endlessly, consuming all available bandwidth.

> **Bridging Loop**: A network condition where redundant paths cause frames to circulate indefinitely, creating broadcast storms and network instability.

Problems Caused by Bridging Loops

Broadcast Storm Example

In a two-switch scenario with parallel links:

- **PC A** connects to **SwA** (port fa0/5)
- **PC B** connects to **SwB** (port fa0/0)
- Two Ethernet cables create a loop between switches

When PC A sends an ARP broadcast to locate PC B:

1. SwA forwards the ARP out of fa0/0 and fa0/1
2. SwB receives the broadcast and forwards it out of remaining ports
3. The process repeats infinitely, creating a **broadcast storm**

Additional Loop-Related Problems

Problem	Description
Multiple Frame Transmission	Unicast frames duplicated, causing protocol errors
MAC-Database Instability	MAC address table flaps as same frame arrives on different ports

Spanning Tree Protocol Core Concepts

STP eliminates loops by logically blocking just enough ports to create a **loop-free topology** while preserving redundancy.

Three Fundamental Steps

1. **Elect one Root Bridge** - the logical center of the tree
2. **Select one Root Port** on every non-root bridge - closest to Root Bridge
3. **Select one Designated Port** on each network segment - forwarding port for that segment

Bridge Protocol Data Unit (BPDU)

Switches exchange **BPDUs** (multicast frames) to share topology information.

Key BPDU Fields

- **Root Bridge ID** - current root's Bridge ID
- **Transmitting Bridge ID** - sender's Bridge ID

Bridge ID Composition

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Bridge ID = Bridge Priority (2 bytes) + MAC Address (6 bytes)
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Example Bridge IDs:

Switch	Bridge Priority	MAC Address	Bridge ID

SwA	32768	0000.0000.9999	32768:0000.0000.999
SwB	32768	0000.0000.1111	32768:0000.0000.111

Root Bridge Election: Lowest Bridge ID wins (priority first, then MAC address). In this example, **SwB** becomes root.

> **Note**: Default priority is **32768**. Administrators can force a switch to become root by lowering priority (must be multiple of 4096).

Cost Metric

STP uses **link-cost** to calculate distance to Root Bridge.

Link Speed	Cost (Revised IEEE)	Cost (Older IEEE)
10 Gbps	2	1
1 Gbps	4	1
100 Mbps	19	10
10 Mbps	100	100

Root Path Cost: Root Bridge = 0, others add link cost to reach root.

Root Port Selection

The **Root Port** receives the **lowest-cost BPDU** from Root Bridge.

Example: Two parallel links between SwA and SwB:

- Upper link (fa0/0) = 10 Mbps → cost 100
- Lower link (fa0/1) = 100 Mbps → cost 19

SwA's fa0/1 becomes Root Port (19 < 100).

Designated Port Selection

STP elects **one Designated Port** per network segment to forward traffic. Other ports become **non-designated (blocked)**.

STP Port States

State	Forward Data?	Learn MAC?	Timer	Type
Blocking	No	No	Max Age = 20s	Stable
Listening	No	No	Forward Delay = 15s	Transitory
Learning	No	Yes	Forward Delay = 15s	Transitory
Forwarding	Yes	Yes	-	Stable

State Details

- **Blocking**: Discards frames (except BPDUs), no MAC learning
- **Listening**: Processes BPDUs, discards frames
- **Learning**: Populates MAC table, still no frame forwarding
- **Forwarding**: Normal operation, frames sent/received

Timer Definitions

- **MaxAge**: Time without BPDUs before topology change (20s)
- **Forward Delay**: Time in Listening/Learning before Forwarding (15s each)

Convergence and Failure Recovery

Convergence: All ports settled into forwarding/blocking states. Default convergence time: **≈50 seconds**.

Failure Recovery Process (when blocked link fails):

1. MaxAge (20s) expires → Blocking to Listening
2. Forward Delay (15s) → Listening to Learning
3. Forward Delay (15s) → Learning to Forwarding **Total**: ≈50 seconds

Section 2: EtherChannel Configuration and Troubleshooting

This section covers EtherChannel technology from a 35-page PDF document (source)

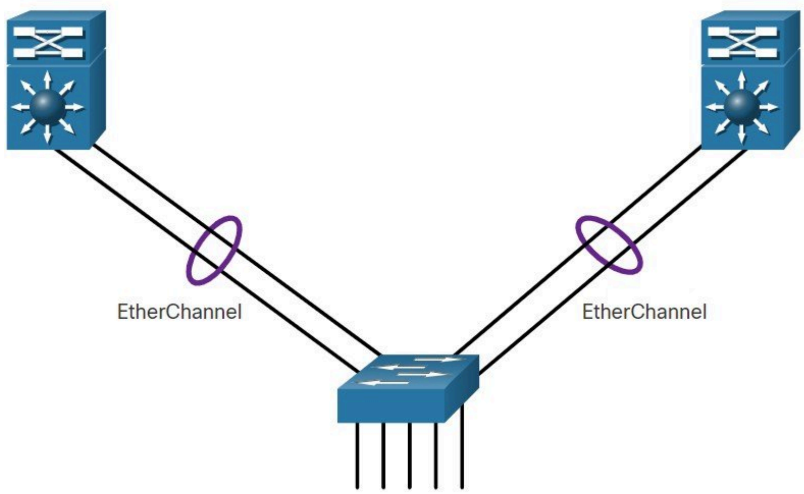
EtherChannel Operation - Link Aggregation

Why EtherChannel?

- Single Ethernet links may not provide sufficient **bandwidth** or **redundancy**
- Adding parallel links helps, but **STP** blocks extra links to prevent loops
- **EtherChannel** allows multiple physical links to act as **one logical link** without STP blocking

Core Definition

EtherChannel = Technology grouping ≥2 physical Ethernet ports into a **single logical interface (port-channel)**



Advantages

Advantage	Explanation
Single-interface configuration	Settings applied to port-channel ensure consistency
No faster media needed	Uses existing switch ports
Load-balancing	Traffic distributed across member links
STP interaction	STP sees **one logical link**, no physical members blocked
Redundancy	Single member loss doesn't affect logical topology

Implementation Restrictions

- **Homogeneous interfaces** - Cannot mix Fast/Gigabit Ethernet
- **Maximum members** - Up to **8** compatible ports per EtherChannel
  - Fast-Ethernet: up to 800 Mbps (full-duplex)
  - Gigabit-Ethernet: up to 8 Gbps
- **Device limits** - Cisco Catalyst 2960 supports **6** simultaneous EtherChannels
- **Configuration consistency** - Both ends must match (speed, duplex, VLAN, trunk)
- **Logical interface** - Port-channel inherits configuration; member changes don't propagate

Auto-Negotiation Protocols

Protocol	Vendor	Standard	Purpose
PAgP	Cisco-proprietary	-	Dynamic EtherChannel on Cisco devices
LACP	IEEE 802.3ad	IEEE	Vendor-neutral dynamic negotiation
Static	-	-	Manual EtherChannel, no negotiation

PAgP - Cisco Proprietary Negotiation

How PAgP Works

- Sends PAgP packets every **30 seconds**
- Checks configuration consistency
- Manages link additions/failures
- Bundles matching links into logical EtherChannel

PAgP Modes

Mode	Behaviour
on	Forces channel without PAgP negotiation
desirable	Active - initiates negotiations

auto	Passive - only responds to PAgP

PAgP Compatibility

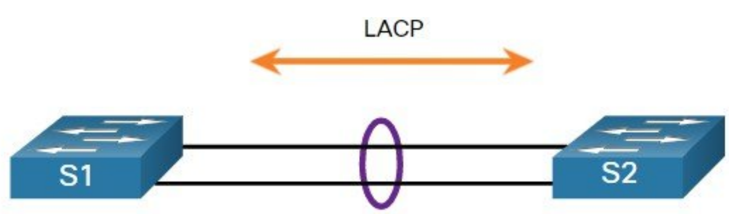
Requirement: Compatible modes needed. Auto+Desirable works, Auto+Auto fails.

S1	S2	Channel?
On	On	✅
On	Desirable/Auto	❌
Desirable	Desirable	✅
Desirable	Auto	✅
Auto	Auto	❌

LACP - IEEE Standard Negotiation

How LACP Works

- Part of **IEEE 802.3ad** (now 802.1AX)
- Sends LACP packets for cross-vendor bundling



LACP Modes

Mode	Behaviour
on	Forces channel, no LACP packets
active	Active - initiates LACP negotiation
passive	Passive - only replies to LACP

LACP Compatibility Table

S1	S2	Channel?
On	On	✅

On	Active/Passive	❌
Active	Active	✅
Active	Passive	✅
Passive	Passive	❌

Configuring EtherChannel

General Guidelines

Guideline	Detail
Support	All interfaces must support EtherChannel
Speed/Duplex	Every member must match
VLAN consistency	Same VLAN or identical trunk settings
Trunking range	Allowed VLAN range must match
Configuration	Apply settings to **port-channel**
Interface type	Can be access, trunk, or routed

LACP Configuration Steps

1. **Select members** - Use interface range command
2. **Create channel** - channel-group mode active
3. **Configure settings** - Enter interface port-channel for settings

Verifying and Troubleshooting

Verification Commands

Command	Purpose
show interfaces port-channel	Port-channel status
show etherchannel summary	One-line channel summary
show etherchannel port-channel	Detailed port-channel view
show interfaces etherchannel	Physical member status

Common Pitfalls

Symptom	Cause
Channel down	Different VLANs or trunk mismatch
Inconsistent native VLAN	Native VLAN mismatch
Partial trunking	Mixed trunk/access settings
VLAN range mismatch	Different allowed VLANs
Mismatched modes	Incompatible negotiation modes

Troubleshooting Workflow

1. **Check summary** - show etherchannel summary
2. **Examine config** - show run | begin interface port-channel
3. **Correct settings** - Fix mismatched modes/settings
4. **Re-verify** - Confirm channel is up

> **STP Note**: When adjusting channels, STP may temporarily block ports. Command order matters.

🎯 Putting it All Together

How Loop Prevention and Link Aggregation Interact

- **STP** blocks excess physical ports to keep the topology loop-free.
- **EtherChannel** bundles those redundant ports into one logical link, so STP sees only a single interface and leaves all members in a forwarding state.
- When a member link fails, the port-channel stays up and traffic continues without invoking STP's 50-second reconvergence cycle.

Core Relationships

> **Spanning Tree Protocol (STP)** – creates a loop-free tree by electing a root bridge, root ports, and designated ports.

> **EtherChannel** – merges ≥ 2 physical interfaces into a single logical port-channel, presenting one entity to STP.

Aspect	STP	EtherChannel
Primary goal	Eliminate bridging loops	Provide bandwidth aggregation & redundancy
Decision point	Block ports to break loops	Keep all member ports forwarding
Failure response	Re-converges (≈ 50 s)	Immediate traffic shift to remaining members
Key config items	Bridge ID, path cost, timers	Speed/duplex, VLAN/trunk consistency, negotiation mode

Practical Implications for Design

- **Use EtherChannel** wherever parallel links are needed; STP will automatically treat the bundle as one link, avoiding blocked ports.
- **Pick one negotiation protocol** (LACP or PAgP) and configure compatible modes on both ends; mismatched modes prevent the channel from forming and force STP to block individual links.
- **Maintain uniform settings** (speed, duplex, VLAN/trunk mode, native VLAN, allowed VLAN list) across all members; any inconsistency breaks the channel and triggers STP blocking.
- **Redundancy planning**: Deploy multiple EtherChannels between core switches. If a loop would arise, STP may block an entire logical link, not a single physical port, preserving a loop-free design while still offering backup capacity.

Quick Checklist for a Healthy STP-EtherChannel Environment

1. **Speed & duplex** – identical on every member interface.
2. **VLAN/trunk settings** – same native VLAN and allowed-VLAN range.
3. **Negotiation protocol** – LACP (active/passive) *or* PAgP (desirable/auto), with compatible modes on both sides.
4. **Verification** – run show etherchannel summary and show spanning-tree to confirm the port-channel appears as a single forwarding link.
5. **Failure test** – disconnect one member; traffic should persist without STP re-convergence.

Interaction Flow (simplified)

1. **Port-channel created** → STP sees one logical port.
2. **Root bridge elected** → All switches compute path costs.

3. **Member link fails** → Port-channel remains up; no new STP calculation needed.
4. **Entire port-channel fails** → STP treats it as a single link loss and may activate a backup logical link, invoking its timers.